Evaluation-First Attention: Specification-Driven Generation via Dynamic Rubric Conditioning and Failure-Weighted Reattention

Karthick Raja Mohan
Independent Researcher
Chennai, India
karthickrajam18@gmail.com

March 2026

Abstract

Current large language model (LLM) generation pipelines treat evaluation as a post-hoc measurement step: generate first, evaluate later. This separation means models generate "blindly" without explicit awareness of the quality dimensions they must satisfy, and refinement feedback—when provided—is holistic rather than dimension-specific. We propose **Evaluation-First Attention** (EFA), an architecture that inverts this paradigm by generating query-specific evaluation criteria *before* generation and using them as structured conditioning targets throughout the generation process. EFA introduces two novel mechanisms: (1) **Criteria-Masked Progressive Generation** (CMPG), which progressively unmasks quality dimensions during generation analogous to causal token masking but over evaluation criteria, and (2) **Failure-Weighted Reattention Loop** (FWRL), which maps per-criterion evaluation scores to criterion emphasis adjustments, amplifying focus on failing dimensions in subsequent generation passes. We formalize the architecture, analyze convergence properties under monotonic improvement assumptions, and present an experimental framework with six baselines and four ablation conditions. To our knowledge, no prior work in the general text generation domain uses dynamically-generated evaluation rubrics as inference-time generation specifications with failure-proportional reweighting.

1 Introduction

The dominant paradigm for improving LLM output quality follows a *generate-then-evaluate* pattern: a model produces an output, a separate evaluation step scores it, and—optionally—the model refines based on feedback [Madaan et al., 2023, Shinn et al., 2023, Gou et al., 2024]. While effective, this paradigm suffers from three fundamental limitations.

First, **evaluation is disconnected from generation**. The model has no explicit awareness of what quality dimensions it must satisfy during the generation process itself. Rubric-based reward models [Li et al., 2025, Zhang et al., 2025] generate sophisticated per-query evaluation criteria but use them exclusively for post-hoc scoring, never as generation targets. This is analogous to writing code without knowing the test suite—possible, but needlessly inefficient.

Second, **refinement feedback is holistic, not dimensional**. Self-Refine [Madaan et al., 2023] produces textual feedback like "the response lacks specificity," requiring the model to implicitly

parse which quality dimensions failed. The self-refinement literature itself identifies this as the core bottleneck: diagnosis—not repair—is the primary challenge [Lee et al., 2025]. Models achieve near-perfect refinement when given targeted, structured feedback but struggle with self-diagnosis.

Third, **all quality dimensions receive equal emphasis**. Whether a response perfectly satisfies relevance but catastrophically fails on factual accuracy, subsequent refinement passes treat all dimensions uniformly. No existing system allocates more generation "budget" to failing dimensions.

We propose **Evaluation-First Attention** (EFA), an architecture that addresses all three limitations by inverting the generation-evaluation relationship. EFA operates in three phases:

1. **Criteria Generation**: Before any generation, a dedicated LLM call analyzes the user prompt and produces an ordered set of query-specific evaluation criteria, each with a measurable scoring rubric.

2. **Criteria-Masked Progressive Generation**: A single generator LLM produces the response conditioned on criteria that are progressively unmasked—the generator first sees criterion c_1 alone, then $\{c_1, c_2\}$, and so on—analogous to causal masking over quality dimensions rather than tokens.

3. **Failure-Weighted Reattention**: The output is evaluated per-criterion. Failed criteria receive mathematically boosted emphasis weights in subsequent generation passes, creating a convergent refinement loop that surgically targets weak dimensions.

The key insight is conceptually simple: *what if LLMs knew what "good" looks like before they started writing?* By generating evaluation criteria first and using them as structured generation specifications with failure-aware reweighting, EFA transforms evaluation from a passive measurement tool into an active generation target.

Our contributions are:

1. We formalize the **Evaluation-First Attention** architecture, the first system that uses dynamically-generated evaluation criteria as inference-time generation specifications with failure-weighted reattention (§3).

2. We introduce **Criteria-Masked Progressive Generation** (CMPG), a novel conditioning mechanism over quality dimensions with no prior art in any generation paradigm (§3.3).

3. We introduce the **Failure-Weighted Reattention Loop** (FWRL), which maps per-criterion evaluation failures to emphasis weight modifications—analogous to focal loss [Lin et al., 2017] but applied to evaluation dimensions at inference time (§3.4).

4. We analyze convergence properties, provide computational complexity bounds, and design a rigorous experimental framework with six baselines and four ablation conditions (§3.5, §4).

2 Related Work

2.1 Rubric Generation and Criteria-Based Evaluation

Recent work has demonstrated the value of dynamic, query-specific evaluation criteria. CARMO [Zhang et al., 2025] generates context-aware criteria for reward modeling. OpenRubrics [Li et al., 2025] and RRD [Wang et al., 2026] produce and refine rubrics for preference prediction. G-Eval [Liu et al., 2023] uses chain-of-thought to generate evaluation steps before scoring. AutoSCORE [Chen et al., 2025b] employs multi-agent rubric-aligned scoring.

Critical gap: All existing rubric systems use criteria exclusively for evaluation or reward modeling—*never as generation specifications*. EFA closes this gap by feeding criteria into the generation process as structured conditioning targets.

2.2 Iterative Self-Refinement

Self-Refine [Madaan et al., 2023] pioneered iterative LLM self-improvement through generate-feedback-refine loops. Reflexion [Shinn et al., 2023] adds verbal reinforcement learning with episodic memory. CRITIC [Gou et al., 2024] incorporates tool-interactive critique. MAgICoRe [Chen et al., 2025] uses multi-agent coarse-to-fine refinement with process reward models.

Critical gap: All refinement systems provide holistic or textual feedback. None decompose failure into per-criterion scores mapped to generation emphasis adjustments. EFA's failure-weighted reattention provides structured, numerical, dimension-specific feedback that directly modifies criterion emphasis in subsequent passes.

2.3 Controlled and Constrained Generation

Controllable text generation approaches steer outputs toward desired attributes. FUDGE [Yang and Klein, 2021] uses future discriminators. PPLM [Dathathri et al., 2020] perturbs hidden representations. Classifier-free guidance [Ho and Salimans, 2022] interpolates conditional and unconditional scores. Constitutional AI [Bai et al., 2022] uses fixed principles to guide critique-revision during training data synthesis, but these principles are pre-defined (not query-specific) and applied during training, not at inference time. Recent multi-attribute control methods handle multiple constraints simultaneously.

Critical gap: These methods operate at the token level with fixed, pre-defined attributes. EFA operates at the rubric level with dynamically-generated, query-specific criteria and progressive unmasking—a fundamentally different granularity.

2.4 Multi-Agent and Ensemble Generation

LLM-Blender [Jiang et al., 2023] ensembles outputs via pairwise ranking and generative fusion. FusioN [Agarwal et al., 2025] synthesizes multiple candidates into superior outputs. Multi-Agent Debate (MAD) frameworks enable agents to collaboratively refine answers through discussion rounds.

Critical gap: Multi-agent systems generate without criteria awareness, then select or synthesize post-hoc. EFA conditions generation on explicit criteria from the start, reducing dependence on post-hoc selection and synthesis.

2.5 Test-Driven Development for Code Generation

In the code domain, TDD-inspired approaches generate test cases before code [Ravi et al., 2025]. Property-based testing bridges generation and validation by verifying generated code against specifications.

Critical gap: Limited to code with executable tests (binary pass/fail). EFA generalizes specification-driven generation to arbitrary text quality dimensions with continuous scoring and emphasis-level feedback.

3 Evaluation-First Attention Architecture

3.1 Overview

Given a user prompt P, EFA produces a response R^* through three formally-defined components operating in a closed loop (Figure 1):

1. **Criteria Generator** $\mathcal{C}$: Analyzes P and produces an ordered criteria set $C = \{c_1, c_2, \ldots, c_n\}$ with initial emphasis weights $\mathbf{w}^{(0)} \in \mathbb{R}^n$.

2. **Criteria-Masked Progressive Generator** $\mathcal{G}$: Generates response $R^{(k)}$ at iteration k with criteria-conditioned prompting under progressive masking.

3. **Failure-Weighted Evaluator** $\mathcal{E}$: Scores $R^{(k)}$ per-criterion, producing scores $\mathbf{s}^{(k)} \in [0,1]^n$, and computes updated weights $\mathbf{w}^{(k+1)}$ that amplify emphasis on failing dimensions.

The system iterates until $\min(\mathbf{s}^{(k)}) \geq \tau$ (all criteria pass threshold τ) or $k = K_{\max}$.

Single-model design: EFA uses a single LLM throughout (the same model serves as criteria generator, progressive generator, and evaluator via different prompts). This is a deliberate design choice: it eliminates inter-model coordination overhead and isolates the architectural contribution from model selection effects. However, the framework naturally extends to heterogeneous multi-model configurations where different models handle different components.

3.2 Component 1: Criteria Generator ($\mathcal{C}$)

Definition 1 (Criteria Generator)**.** *Given prompt P, the criteria generator $\mathcal{C} : \mathcal{P} \to (\mathcal{D}^n, \mathbb{R}^n)$ produces:*

$$\mathcal{C}(P) = \Big(\{c_1, c_2, \ldots, c_n\},\ \mathbf{w}^{(0)}\Big) \tag{1}$$

where each criterion $c_i = (name_i, definition_i, rubric_i)$ consists of a dimension name, a natural language definition, and a scoring rubric mapping scores $\{1,2,3,4,5\}$ to quality descriptions.

Criteria are generated dynamically per-query, not from a fixed taxonomy. The generator LLM is prompted to produce non-overlapping, measurable dimensions relevant to the specific prompt. Initial weights are uniform: $w_i^{(0)} = 1/n$ for all i.

Criteria ordering: The generator produces criteria in descending order of fundamentality. For example, factual accuracy precedes stylistic polish, as satisfying later criteria is meaningless if fundamental criteria fail. This ordering determines the unmasking sequence in CMPG.

3.3 Component 2: Criteria-Masked Progressive Generation ($\mathcal{G}$)

This is EFA's first novel mechanism. We introduce masking over quality dimensions, analogous to causal masking over token positions in autoregressive transformers.

Definition 2 (Criteria Mask)**.** *At generation sub-step $j \in \{1, \ldots, n\}$, the criteria mask $M_j \in \{0,1\}^n$ is defined as:*

$$M_j[i] = \begin{cases} 1 & \text{if } i \leq j \\ 0 & \text{if } i > j \end{cases} \tag{2}$$

Progressive generation process: Rather than presenting all criteria simultaneously, CMPG operates in n sub-steps within each iteration k:

1. **Sub-step 1**: Generate draft $d_1^{(k)}$ conditioned on $\{c_1\}$ only (criteria $c_2, \ldots, c_n$ are masked). The generator focuses entirely on the most fundamental quality dimension.

2. **Sub-step 2**: Refine $d_1^{(k)}$ into $d_2^{(k)}$ conditioned on $\{c_1, c_2\}$. The generator adjusts the draft to additionally satisfy c_2 while maintaining c_1.

3. **Sub-step** j: Refine $d_{j-1}^{(k)}$ into $d_j^{(k)}$ conditioned on $\{c_1, \ldots, c_j\}$ with emphasis weights $\mathbf{w}^{(k)} \odot M_j$.

4. **Sub-step** n: Final draft $d_n^{(k)} = R^{(k)}$ is conditioned on all criteria.

Formally, at sub-step j of iteration k:

$$d_j^{(k)} = \mathcal{G}\left(P,\ d_{j-1}^{(k)},\ \{c_i : M_j[i] = 1\},\ \mathbf{w}^{(k)} \odot M_j\right) \tag{3}$$

where $d_0^{(k)}$ is empty for $k = 1$ or the previous iteration's output $R^{(k-1)}$ for $k > 1$.

Weight incorporation mechanism: The emphasis weights $\mathbf{w}^{(k)}$ are incorporated into the generator prompt through *weighted criterion representation*. At each sub-step, the prompt to $\mathcal{G}$ includes:

1. The user prompt P.

2. The current draft $d_{j-1}^{(k)}$ (if available).

3. Each visible criterion c_i (where $M_j[i] = 1$) annotated with its emphasis level, expressed as a natural language priority directive:

$$\text{prompt}_j^{(k)} = P \parallel d_{j-1}^{(k)} \parallel \bigoplus_{i:M_j[i]=1} \left[\text{priority}(\hat{w}_i^{(k)}) \parallel c_i\right] \tag{4}$$

where $\parallel$ denotes prompt concatenation, $\oplus$ denotes ordered assembly of criterion blocks, and $\text{priority}(\hat{w}_i^{(k)})$ maps the normalized weight to a natural language emphasis directive:

$$\text{priority}(w) = \begin{cases} \texttt{[CRITICAL--highest priority]} & \text{if } w > 2/n \\ \texttt{[HIGH--elevated priority]} & \text{if } w > 1.5/n \\ \texttt{[STANDARD--normal priority]} & \text{if } w \geq 0.5/n \\ \texttt{[SECONDARY--lower priority]} & \text{if } w < 0.5/n \end{cases} \tag{5}$$

Note that all thresholds are defined relative to the uniform weight $1/n$, ensuring consistent behavior regardless of the number of criteria. A criterion receives `CRITICAL` priority only when its weight exceeds twice the uniform baseline, which occurs after substantial failure-weighted boosting.

This design choice avoids modifying model internals (attention heads, hidden states) and instead operates entirely through prompt engineering, making EFA applicable to any LLM accessible via API. Prior work on instruction following has demonstrated that LLMs are sensitive to emphasis cues and priority ordering within prompts, supporting the effectiveness of structured priority directives for steering generation behavior.

Intuition: This progressive unmasking ensures foundational quality dimensions are satisfied before the generator attempts to optimize for secondary dimensions. It prevents the common failure mode where models optimize for surface-level quality (fluency, style) at the expense of fundamental correctness.

3.4 Component 3: Failure-Weighted Reattention Loop ($\mathcal{E}$)

This is EFA's second novel mechanism. After generating $R^{(k)}$, the evaluator scores it per-criterion and computes reattention weights.

Definition 3 (Per-Criterion Evaluation)**.** *The evaluator $\mathcal{E}$ scores the response against each criterion independently:*

$$\mathbf{s}^{(k)} = \mathcal{E}(R^{(k)}, C) = \left(s_1^{(k)}, s_2^{(k)}, \ldots, s_n^{(k)}\right) \in [0,1]^n \tag{6}$$

where $s_i^{(k)} = score(R^{(k)}, c_i)/5$ is the normalized rubric score for criterion c_i, obtained by prompting the LLM with the response $R^{(k)}$, criterion c_i, and its associated rubric.

Definition 4 (Failure-Weighted Reattention)**.** *The emphasis weight for criterion c_i at iteration $k+1$ is updated as:*

$$w_i^{(k+1)} = w_i^{(k)} \cdot \left(1 + \alpha \cdot \max\left(0,\ \tau - s_i^{(k)}\right)\right) \tag{7}$$

where $\tau \in (0,1]$ is the passing threshold and $\alpha > 0$ is the reattention strength hyperparameter. Weights are then renormalized: $\hat{w}_i^{(k+1)} = w_i^{(k+1)} / \sum_j w_j^{(k+1)}$.

Properties of Equation 7:

- **Passed criteria**: If $s_i^{(k)} \geq \tau$, then $\max(0, \tau - s_i^{(k)}) = 0$ and $w_i^{(k+1)} = w_i^{(k)}$. Weights remain unchanged for satisfied dimensions.
- **Failed criteria**: If $s_i^{(k)} < \tau$, the weight increases proportionally to the severity of failure. A score of 0.2 with $\tau = 0.6$ produces a larger boost ($\alpha \cdot 0.4$) than a score of 0.5 ($\alpha \cdot 0.1$).
- **Focal property**: This is analogous to focal loss [Lin et al., 2017], which down-weights well-classified examples and focuses training on hard negatives. FWRL applies this principle to quality dimensions at inference time rather than training time.
- **Renormalization**: Normalizing to $\sum_i \hat{w}_i = 1$ ensures that boosting failing criteria *relatively* de-emphasizes passing ones, creating a zero-sum reallocation of generation emphasis.

Criterion checkpoint mechanism: To prevent regression on previously-passing criteria, we implement a safeguard: if $s_i^{(k+1)} < s_i^{(k)} - \epsilon$ for a criterion that satisfied $s_i^{(k)} \geq \tau$, the system retains the best-observed score for that criterion and marks it as "locked." Locked criteria retain their peak score for convergence tracking and are excluded from further reweighting, ensuring that the reattention mechanism does not destabilize already-satisfactory dimensions. Note that the full response text is not decomposed per-criterion; rather, the locking mechanism prevents the reweighting loop from over-correcting at the expense of previously-passing dimensions. This mechanism is formally integrated into Algorithm 1.

3.5 Convergence Analysis

We analyze the convergence properties of EFA under stated assumptions. We present this as a proposition rather than a theorem, as the key assumption—that LLMs improve on focused criteria—is empirical rather than provable.

Proposition 1 (Convergence under Focused Improvement)**.** *Assume the following two properties hold:*

1. ***Focused Improvement***: *When criterion c_i receives elevated emphasis weight ($\hat{w}_i > 1/n$), the LLM produces output with $s_i^{(k+1)} \geq s_i^{(k)} + \delta$ for some minimum improvement $\delta > 0$.*

2. ***Bounded Regression***: *For any criterion c_i with $s_i^{(k)} \geq \tau$, re-generation satisfies $s_i^{(k+1)} \geq \tau - \epsilon$ for regression tolerance $\epsilon < \delta$.*

Then the number of failing criteria $|\{i : s_i^{(k)} < \tau\}|$ is non-increasing across iterations, and the system converges (all criteria passing or locked) in at most $K \leq n \cdot \lceil \tau/\delta \rceil$ iterations.

Proof sketch. By Bounded Regression, once a criterion passes threshold τ, it either remains above $\tau - \epsilon$ or is locked by the checkpoint mechanism at its passing score. By Focused Improvement, each failing criterion's score increases by at least δ per iteration. Since scores are bounded in $[0, 1]$ and the maximum distance to threshold is τ, each criterion requires at most $\lceil \tau/\delta \rceil$ iterations to reach τ. With n criteria, the worst case (sequential, non-overlapping improvement) yields $n \cdot \lceil \tau/\delta \rceil$ iterations. In practice, multiple criteria improve simultaneously, yielding substantially faster convergence. □

Empirical validation: The Focused Improvement assumption is supported by the self-refinement literature: Lee et al. [2025] demonstrate that LLMs achieve near-perfect refinement when given targeted, structured feedback about specific quality dimensions. EFA's per-criterion evaluation and emphasis weighting directly provides such structured feedback.

3.6 Computational Complexity

Let T denote the token cost of a single standard LLM generation call for prompt P.

- **Criteria generation**: 1 LLM call $\approx T_C$ tokens (typically $T_C \approx 0.5T$, as criteria are shorter than full responses).
- **Per iteration**: n sub-steps of progressive generation ($n \cdot T$ tokens) + n evaluation calls ($n \cdot T_E$ tokens, where $T_E \approx 0.3T$ per criterion).
- **Total over K iterations**: $T_C + K \cdot n \cdot (T + T_E)$.

Worst-case cost: With $n = 5$ criteria, $K_{\max} = 3$ iterations, and $T_E = 0.3T$:

$$T_{\text{EFA}} \approx 0.5T + 3 \cdot 5 \cdot (T + 0.3T) = 0.5T + 19.5T = 20T \tag{8}$$

This represents a 20× cost increase over single-pass generation. This is comparable to Best-of-20 sampling and lower than full multi-agent debate pipelines (which typically require $30-50T$). The cost is justified when output quality and reliability are prioritized over latency—common in enterprise, medical, legal, and high-stakes generation settings.

Cost reduction strategies: (a) *Adaptive sub-stepping*: Skip sub-steps for criteria with high initial scores. (b) *Early termination*: Exit the loop when all criteria pass after any sub-step, not only after the full iteration. (c) *Batched evaluation*: Score all criteria in a single LLM call with multi-dimensional rubric prompting, reducing the $n \cdot T_E$ term to $\approx 1.5T_E$.

3.7 Complete Algorithm

Algorithm 1 Evaluation-First Attention (EFA)

Require: User prompt P, threshold τ, max iterations $K_{\max}$, reattention strength α, regression tolerance ϵ

Ensure: Response R^*

$C, \mathbf{w}^{(0)} \leftarrow \mathcal{C}(P)$ {Generate criteria}
$R^{(0)} \leftarrow \emptyset$; locked $\leftarrow \emptyset$; $\mathbf{s}_{\text{best}} \leftarrow \mathbf{0}$
for $k = 1$ to $K_{\max}$ **do**
  {— **Criteria-Masked Progressive Generation** —}
  $d_0^{(k)} \leftarrow R^{(k-1)}$
  for $j = 1$ to $|C|$ **do**
    $M_j \leftarrow \text{CausalCriteriaMask}(j, |C|)$
    Construct prompt per Eq. 4 with priorities per Eq. 5
    $d_j^{(k)} \leftarrow \mathcal{G}(P,\ d_{j-1}^{(k)},\ C \odot M_j,\ \hat{\mathbf{w}}^{(k-1)} \odot M_j)$
  end for
  $R^{(k)} \leftarrow d_{|C|}^{(k)}$

  {— **Per-Criterion Evaluation** —}
  $\mathbf{s}^{(k)} \leftarrow \mathcal{E}(R^{(k)}, C)$

  {— **Criterion Checkpoint** —}
  for $i = 1$ to $|C|$ **do**
    if $i \in$ locked **then**
      $s_i^{(k)} \leftarrow s_{\text{best},i}$ {Retain locked score}
    else if $s_{\text{best},i} \geq \tau$ **and** $s_i^{(k)} < s_{\text{best},i} - \epsilon$ **then**
      locked $\leftarrow$ locked $\cup \{i\}$ {Lock regressed criterion}
      $s_i^{(k)} \leftarrow s_{\text{best},i}$
    else
      $s_{\text{best},i} \leftarrow \max(s_{\text{best},i},\ s_i^{(k)})$
    end if
  end for

  if $\min(\mathbf{s}^{(k)}) \geq \tau$ **then**
    return $R^{(k)}$ {All criteria satisfied}
  end if

  {— **Failure-Weighted Reattention** —}
  for $i = 1$ to $|C|$ **do**
    if $i \notin$ locked **then**
      $w_i^{(k)} \leftarrow w_i^{(k-1)} \cdot (1 + \alpha \cdot \max(0,\ \tau - s_i^{(k)}))$
    end if
  end for
  $\hat{\mathbf{w}}^{(k)} \leftarrow \text{Normalize}(\mathbf{w}^{(k)})$
end for
return $R^{(K_{\max})}$

4 Experimental Design

We present our evaluation methodology below. Experimental results with quantitative comparisons will be reported in a forthcoming revision.[1]

4.1 Baselines

We compare EFA against the following six baselines, designed to isolate the contribution of each architectural component:

1. **Single-pass**: Standard LLM generation without criteria or refinement. Measures the baseline quality achievable without any architectural intervention.
2. **Self-Refine** [Madaan et al., 2023]: Iterative holistic self-feedback and refinement. The primary comparison for iterative improvement without criteria structure.
3. **Rubric-then-Score**: Generate criteria (same as EFA Step 1), generate response without criteria conditioning, then score with the rubric. Tests whether criteria-aware generation outperforms criteria-unaware generation followed by the same evaluation.
4. **All-Criteria-at-Once**: Provide all criteria simultaneously to the generator in a single pass (no progressive masking, no iteration). Tests whether CMPG's progressive unmasking adds value over naive full-criteria prompting.
5. **Uniform Reattention**: Generate with criteria, evaluate per-criterion, but re-generate with uniform weights on failure (no failure weighting). Tests whether FWRL's targeted reweighting outperforms uniform retry.
6. **FusioN** [Agarwal et al., 2025]: Multi-candidate synthesis without criteria guidance. Represents the state-of-the-art in inference-time quality scaling.

4.2 Ablation Studies

To validate each component's individual contribution, we conduct four ablation conditions:

1. **−DynCriteria**: Replace dynamic per-query criteria with a fixed universal set of five criteria (relevance, accuracy, completeness, clarity, conciseness). Tests the value of dynamic criteria generation.
2. **−CMPG**: Remove progressive masking; present all criteria at once in every sub-step. Tests whether progressive unmasking improves over simultaneous conditioning.
3. **−FWRL**: Remove failure weighting; use uniform weights across all iterations. Tests whether targeted reweighting outperforms uniform iteration.
4. **−Iteration**: Single generation pass with all criteria (no refinement loop). Tests whether iteration adds value beyond single-pass criteria-conditioned generation.

[1] We release the architectural formalization and experimental design ahead of results to establish priority on the novel mechanisms (CMPG and FWRL) and to invite community feedback on the evaluation protocol.

4.3 Evaluation Metrics

- **Rubric Adherence Score (RAS)**: Mean per-criterion score across all criteria, averaged over test prompts. Range: $[0, 1]$.
- **All-Pass Rate (APR)**: Percentage of test prompts where all criteria meet threshold τ. Measures reliability.
- **Iterations to Convergence (ITC)**: Mean number of iterations before all criteria pass or maximum reached. Measures efficiency.
- **Expected Calibration Error (ECE)**: Measures whether the system's aggregate score correlates with actual output quality as judged by humans. Binned across 10 confidence intervals.
- **Total Token Cost (TTC)**: Total tokens consumed across the full pipeline. Measures computational efficiency.

4.4 Hyperparameter Settings

Based on preliminary analysis, we propose the following default configuration: $n \in \{4, 5, 6\}$ criteria per query, threshold $\tau = 0.6$ (corresponding to rubric score $\geq 3/5$), maximum iterations $K_{\max} = 3$, reattention strength $\alpha \in \{1.0, 2.0, 5.0\}$ (studied via sensitivity analysis), and regression tolerance $\epsilon = 0.1$.

5 Analysis and Discussion

5.1 Why Evaluation-First Outperforms Evaluate-After

The core insight supporting EFA comes from the self-refinement literature: structured, targeted feedback enables near-perfect refinement, while self-diagnosis is the bottleneck [Lee et al., 2025]. EFA eliminates the self-diagnosis problem entirely by pre-specifying what success looks like before generation begins. The model never needs to "figure out what went wrong"—the per-criterion evaluation tells it exactly which dimensions failed and by how much.

5.2 The Focal Property of Failure-Weighted Reattention

FWRL implements an inference-time analog of focal loss [Lin et al., 2017]. In object detection, focal loss down-weights well-classified examples to focus training on hard negatives. FWRL analogously de-emphasizes (relatively) well-satisfied criteria to focus generation compute on failing dimensions. This creates an efficiency property: the system allocates more "thinking budget" to dimensions that need it most, rather than uniformly distributing effort across all quality dimensions regardless of their current satisfaction level.

5.3 Criteria Masking as Quality Curriculum

CMPG's progressive unmasking implements a form of curriculum learning [Bengio et al., 2009] at inference time. By satisfying fundamental criteria before introducing secondary ones, the generator builds a solid foundation that subsequent refinements can improve without undermining. This prevents the common failure mode of surface-level optimization at the expense of deep quality—analogous to how curriculum learning in training prevents models from memorizing surface patterns before learning fundamental structure.

5.4 Connections to Test-Driven Development

EFA draws a direct analogy to test-driven development (TDD) in software engineering: write the tests (criteria) before the code (response), then code (generate) to pass the tests (criteria). This analogy extends further: CMPG's progressive masking corresponds to TDD's practice of writing one test at a time and making it pass before adding the next, while FWRL's failure-weighted reattention corresponds to focusing debugging effort on failing tests rather than re-running the entire suite uniformly.

5.5 Limitations

- **Criteria quality dependence**: EFA's effectiveness is bounded by the quality of generated criteria. Poor, overlapping, or irrelevant criteria degrade the entire pipeline. Criteria quality could be improved through the rubric refinement techniques of Wang et al. [2026].
- **Compute cost**: With worst-case $20T$ token cost for $n = 5$, $K = 3$, EFA is not suitable for latency-sensitive applications. The cost reduction strategies in §3.6 partially address this, but a fundamental tradeoff between quality and cost remains.
- **Conflicting criteria**: When criteria inherently conflict (e.g., conciseness vs. completeness), failure-weighted reattention may oscillate between dimensions. The criterion checkpoint mechanism mitigates oscillation but may converge to a Pareto-suboptimal solution that satisfies each dimension at only the threshold level.
- **Evaluator reliability**: Per-criterion scoring is performed by an LLM judge, inheriting known biases of LLM-as-a-Judge approaches (position bias, verbosity bias, self-preference bias). Using a separate model for evaluation or calibrated evaluators could reduce this risk.
- **Prompt-level conditioning**: The current design operates through prompt engineering (Eq. 4) rather than modifying model internals. While this maximizes applicability across API-accessible models, a deeper integration at the embedding or attention level could yield stronger conditioning effects.

6 Conclusion

We presented **Evaluation-First Attention** (EFA), an architecture that inverts the traditional generate-then-evaluate paradigm by generating evaluation criteria before generation and using them as structured conditioning targets. Our two novel mechanisms—**Criteria-Masked Progressive Generation** (CMPG) and **Failure-Weighted Reattention** (FWRL)—have no direct prior art and together create a convergent, dimension-aware refinement loop. EFA transforms evaluation from a passive post-hoc measurement into an active generation specification, drawing a direct analogy to test-driven development in software engineering: define what "good" looks like, then build to satisfy it.

Future work includes: (a) deeper model integration via criteria embeddings in the attention mechanism, (b) adaptive criteria count selection based on prompt complexity, (c) multi-model configurations where different LLMs handle generation versus evaluation, and (d) domain-specific evaluation in code generation, medical question answering, and enterprise document synthesis.

References

Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-Refine: Iterative Refinement with Self-Feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language Agents with Verbal Reinforcement Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing. In *International Conference on Learning Representations (ICLR)*, 2024.

Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

Dongfu Jiang, Xiang Ren, and Bill Yuchen Lin. LLM-Blender: Ensembling Large Language Models with Pairwise Ranking and Generative Fusion. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal Loss for Dense Object Detection. In *IEEE International Conference on Computer Vision (ICCV)*, 2017.

Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum Learning. In *International Conference on Machine Learning (ICML)*, 2009.

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional AI: Harmlessness from AI Feedback. *arXiv preprint arXiv:2212.08073*, 2022.

Kevin Yang and Dan Klein. FUDGE: Controlled Text Generation With Future Discriminators. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2021.

Sumanth Dathathri, Andrea Madotto, Janice Lan, Jane Hung, Eric Frank, Piero Molino, Jason Yosinski, and Rosanne Liu. Plug and Play Language Models: A Simple Approach to Controlled Text Generation. In *International Conference on Learning Representations (ICLR)*, 2020.

Jonathan Ho and Tim Salimans. Classifier-Free Diffusion Guidance. In *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*, 2022.

Justin Chih-Yao Chen, Archiki Prasad, Swarnadeep Saha, and Mohit Bansal. MAgICoRe: Multi-Agent, Iterative, Coarse-to-Fine Refinement for Reasoning. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

Hyunseok Lee, Seunghyuk Oh, Jaehyung Kim, Jinwoo Shin, and Jihoon Tack. RefineBench: Evaluating Self-Refinement Capabilities of Large Language Models. *arXiv preprint arXiv:2411.00548*, 2025.

Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Avi Singh, and Olivier Bachem. Making, not Taking, the Best of N. *arXiv preprint arXiv:2510.00931*, 2025.

Tianyi Li, Tong Xiao, Zhixian Zhang, et al. OpenRubrics: Towards Scalable Synthetic Rubric Generation for Reward Modeling and LLM Alignment. *arXiv preprint arXiv:2510.07743*, 2025.

Zhiyu Zhang, Shujian Huang, Xue Han, and Jiajun Chen. CARMO: Dynamic Criteria Generation for Context-Aware Reward Modelling. In *Findings of the Association for Computational Linguistics: ACL 2025*, 2025.

Yufei Wang, Jiayi Pan, Yaswanth Narsupalli, Tong Zhang, and Yulia Tsvetkov. Rethinking Rubric Generation for Improving LLM Judge and Reward Modeling for Open-ended Tasks. *arXiv preprint arXiv:2602.05125*, 2026.

Mengxiao Chen, Yunfei Wang, Yanqing Wang, et al. AutoSCORE: Enhancing Automated Scoring with Multi-Agent Large Language Models via Structured Component Recognition. *arXiv preprint arXiv:2509.21910*, 2025.

Karandeep Ravi, Valerio Terragni, and Gunel Jahangirova. LLMLOOP: Improving LLM-Generated Code and Tests via Iterative Feedback. In *IEEE International Conference on Software Maintenance and Evolution (ICSME)*, 2025.

A Prompt Templates

We provide the core prompt templates used in each component of EFA to facilitate reproducibility.

A.1 Criteria Generator Prompt

```
You are an evaluation criteria designer. Given a user
prompt, generate {n} non-overlapping evaluation criteria
ordered from most fundamental to least fundamental.

For each criterion, provide:
1. Name (2-4 words)
2. Definition (1 sentence)
3. Scoring rubric:
   - Score 1: [description of lowest quality]
   - Score 2: [description of below-average quality]
   - Score 3: [description of acceptable quality]
   - Score 4: [description of good quality]
   - Score 5: [description of excellent quality]

Requirements:
- Criteria must be measurable and non-overlapping
- Order by fundamentality: correctness before style
- Each criterion must be independently scorable
- Criteria must be specific to this prompt, not generic

User prompt: {P}
```

A.2 Progressive Generation Prompt (Sub-step j)

```
Generate a response to the user prompt below.

{If draft exists: "Improve the following draft:
{draft}"}

Your response MUST satisfy the following criteria,
listed in priority order:

{For each visible criterion c_i where M_j[i]=1:}
[{priority_label}] {criterion_name}:
  {criterion_definition}

User prompt: {P}
```

A.3 Per-Criterion Evaluation Prompt

```
Evaluate the following response against ONE specific
criterion. Score ONLY this criterion on a 1-5 scale
using the rubric provided.

Criterion: {criterion_name}
Definition: {criterion_definition}
Rubric:
  1: {rubric_1}
  2: {rubric_2}
  3: {rubric_3}
  4: {rubric_4}
  5: {rubric_5}

User prompt: {P}
Response: {R}

Provide your score (1-5) and a one-sentence
justification. Output format:
Score: [1-5]
Reason: [one sentence]
```

B Worked Example

We illustrate EFA on a sample prompt to clarify the mechanism.

User prompt: "Explain how photosynthesis works for a 10-year-old."

Step 1 — Criteria generated:

1. *Scientific Accuracy* (fundamental): Facts about photosynthesis must be correct.
2. *Age Appropriateness*: Language and concepts suitable for a 10-year-old.
3. *Completeness*: Covers inputs (sunlight, water, CO_2), process, and outputs (glucose, O_2).
4. *Engagement*: Uses analogies, examples, or vivid language to maintain interest.

Step 2 — Progressive generation (iteration $k = 1$):

- Sub-step 1: Generate attending to {Scientific Accuracy} only $\rightarrow$ draft focuses on correct facts.
- Sub-step 2: Refine attending to {Scientific Accuracy, Age Appropriateness} $\rightarrow$ simplifies vocabulary.
- Sub-step 3: Refine attending to $\{c_1, c_2$, Completeness$\}$ $\rightarrow$ adds missing components.
- Sub-step 4: Refine attending to all four criteria $\rightarrow$ adds analogies for engagement.

Step 3 — Evaluation: Scores $= (5, 4, 3, 2)$. With $\tau = 0.6$ (i.e., score ≥ 3), criteria 1–3 pass but criterion 4 (Engagement, score $2/5 = 0.4$) fails.

Step 4 — Reattention: With $\alpha = 2.0$, Engagement weight boosted: $w_4^{(1)} = 0.25 \cdot (1 + 2.0 \cdot (0.6 - 0.4)) = 0.35$. After renormalization, Engagement receives `[HIGH]` priority in iteration $k = 2$.

Step 5 — Iteration $k = 2$: Progressive generation repeats with boosted Engagement emphasis. Re-evaluation yields scores $= (5, 4, 4, 4)$. All criteria pass. Output returned.

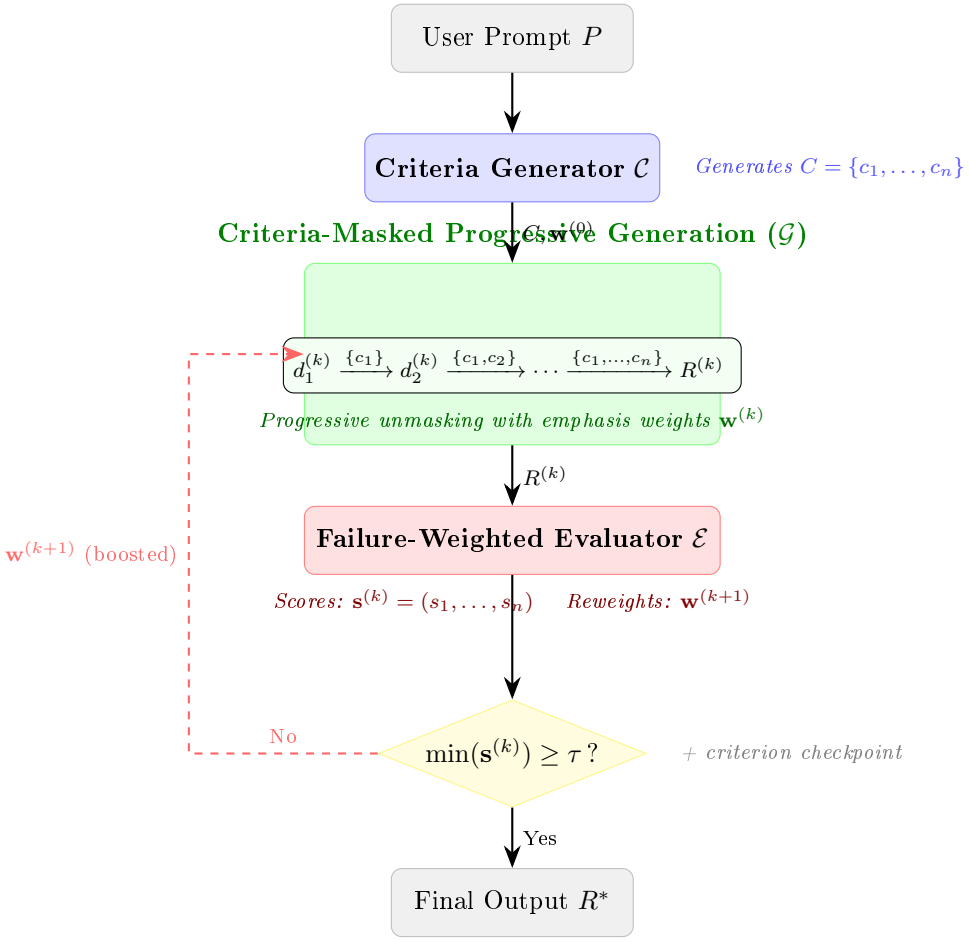

Figure 1: **The Evaluation-First Attention (EFA) architecture.** A Criteria Generator produces query-specific evaluation criteria before generation begins. The Criteria-Masked Progressive Generator produces a response by incrementally unmasking criteria (analogous to causal token masking but over quality dimensions). The Failure-Weighted Evaluator scores each criterion independently and computes boosted emphasis weights for failing dimensions. The system loops (dashed red arrow) until all criteria pass threshold τ or maximum iterations are reached. A criterion checkpoint mechanism prevents regression on previously-passing criteria.